Ternary codes from the Witt design and the plane of order four

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Abstract

We derive the ternary graph codes studied by Leemans and Rodrigues from one incidence calculation and the common criteria for finite module and automorphism certificates. The complete weight records and actual permutation actions determine all four principal codes, their specified submodules and full coordinate automorphism groups. We also give the prime-rank table, the socle and auxiliary-module calculations, and the precise qualifications needed for several ancillary source statements. Classifications of other irreducible representations remain explicitly retained inputs, distinct from the irreducibility certificates for the modules constructed here.

The source is the complete eleven-page 2016 edition of Leemans–Rodrigues [1]. All four clauses of its main Theorem 1 are addressed, together with Lemmas 1–2, Propositions 1–5, Corollary 1, Tables 1–2 and the specified modules and orbits in Remarks 1–3. We give the corrected meaning of the ancillary assertions that cannot hold literally. The common paper [2] supplies full proofs of the projection, incidence, matrix-algebra, socle, refinement and weight identities used below. Its whole cost belongs to the topic; it is not a free import from another application. Every finite certificate is bound to the displayed construction and its actual basis. The programme's numerical comparisons and source admissions are separate from this draft.

1 The four graph codes

Throughout, automorphisms of a code mean permutations of its coordinates, not arbitrary monomial transformations. Unless a prime is displayed, the field is $F = \mathbb{F}_3$. The constant vector is denoted by j, with its length determined by context. A direct sum of submodules below is an internal sum in the same coordinate space, rather than a concatenation of shorter codes. For a symmetric adjacency matrix A_n of order n, put

$$C_n = \operatorname{im} A_n, \qquad D_n = \langle (A_n)_i - (A_n)_1 : 1 \le i \le n \rangle.$$

For length 120 we also write $S = D_{120}$, to agree with the source. The code of the complementary graph is denoted by $\overline{C}_n$. This notation distinguishes a graph complement from an orthogonal dual.

The main results of Leemans and Rodrigues [1] concern four specified strongly regular graphs. We use their graph constructions and the identifications with the relevant McLaughlin subgraphs, as attributed in that paper to Brouwer–van Eijl and the preceding graph literature. The actual graphs are fixed here, so assertions about a particular code are not inferred merely from four strongly regular parameters.

For $n = 105$, use the incident point–line flags of $\mathrm{PG}(2, 4)$. Join distinct flags $(p, \ell), (q, m)$ if $p = q$, or $\ell = m$, or neither cross incidence $p \in m$, $q \in \ell$ holds. The finite field is $\mathbb{F}_4 = \mathbb{F}_2[z]/(z^2 + z + 1)$. All 105 vertices and their adjacencies are constructed directly from this field. The printed adjacency sentence on page 242 has a defect: a conjunctive reading gives valency eight, while joining all its clauses disjunctively gives forty. Our identification with the intended graph therefore retains Coolsaet's uniqueness theorem for the complementary

parameters $(105, 32, 4, 12)$ [3, Theorem 3]; the verified complementary parameters alone would not identify a graph without that input.

For the other three lengths, let $\mathcal{D}$ be the specified $S(4, 7, 23)$ with its 253 heptads. One complete construction takes the weight-seven words in the binary span of the twelve shifts of

$$1 + x + x^5 + x^6 + x^7 + x^9 + x^{11}.$$

The supplied constructor verifies all 4096 words, their weight multiplicities, and the unique containment of every four-subset in a heptad. It also verifies all block intersection sizes. Delete a fixed set of t points, where $t = 0, 1, 2$, and retain the heptads avoiding it. There are respectively $253, 176, 120$ such blocks. Join distinct retained blocks when their intersection has size three. This is exactly the complement of the source's intersection-one graph. With H_t the point–block incidence matrix on the remaining $23 - t$ points, the common paper [2, Theorem 2.2] gives

t	n	$H_t H_t^{\mathsf{T}} \pmod 3$	C_n
0	253	$2I_{23}$	$\operatorname{im} H_0^{\mathsf{T}}$
1	176	$I_{22} + J_{22}$	$H_1^{\mathsf{T}}(e^{\perp})$
2	120	I_{21}	$\operatorname{im} H_2^{\mathsf{T}}$.

(1)

Here e is the constant point vector. Each of these three codes is LCD; $\overline{C}_n = C_n^{\perp}$ for $n = 120, 176$. Also $C_{253} = Fj \perp\!\!\!\oplus D_{253}$, with $D_{253} = H_0^{\mathsf{T}}(e^{\perp})$ of dimension 22 and LCD. The hypotheses of the common theorem are discharged by the complete Steiner and intersection checks, not by assuming the ternary ranks.

The verified parameters and integral spectra are

n	(n, k, λ, μ)	spectrum of A_n
105	$(105, 72, 51, 45)$	$72^1, 9^{20}, (-3)^{84}$
120	$(120, 77, 52, 44)$	$77^1, 11^{20}, (-3)^{99}$
176	$(176, 105, 68, 54)$	$105^1, 17^{21}, (-3)^{154}$
253	$(253, 140, 87, 65)$	$140^1, 25^{22}, (-3)^{230}$.

(2)

The finite certificate checks every vertex degree and every pair's common-neighbour count. The spectra then follow from $A^2 = (\lambda-\mu)A+(k-\mu)I+\mu J$, the constant eigenvector, and the trace and dimension identities. Connectivity follows from $\mu > 0$, so the constant eigenvalue has multiplicity one.

1.1 The retained orbital construction

The source's Result 1 attributes a general construction to Key and Moori, including their correction. Its argument is short enough to make the exact scope explicit. Let G act primitively on Ω, let $\alpha \in \Omega$, and let $\Delta \neq \{\alpha\}$ be a nonempty orbit of G_α. The set stabilizer G_Δ contains G_α. Primitivity makes the latter maximal; since Δ is a proper nonempty subset, $G_\Delta \neq G$. Hence $G_\Delta = G_\alpha$, and the translates of Δ are $|\Omega|$ distinct blocks. Transitivity and counting incidences give constant block size and replication $|\Delta|$, producing the symmetric 1-design. If the orbital is self-paired, these blocks are precisely the neighbour sets of the associated undirected graph. The nonempty graph's components form a G-invariant partition, so it is connected. Its symmetric incidence matrix identifies the design with its dual, and both point and block actions are primitive. This preserves the background result without using it in place of any of the explicit constructions above.

2 Ranks and the complete finite weight records

The two prime-rank lemmas in the source, and its surrounding assertion about other characteristics, can be handled together. For the fixed matrices the full answer is

n	exceptional primes and ranks	rank at every other prime
105	$2:104,\quad 3:20$	105
120	$3:21,\quad 7:119,\quad 11:99$	120
176	$3:21,\quad 5:175,\quad 7:175,\quad 17:155$	176
253	$2:252,\quad 3:23,\quad 5:230,\quad 7:252$	253.

(3)

Indeed (2) determines the integer determinant as the product of the listed eigenvalues with their multiplicities. Outside its prime divisors reduction is invertible. For every prime divisor, the complete supplied elimination checks the rank of the actual matrix. Thus the finite table covers every exceptional characteristic; no sample of small primes is being extrapolated. For the ternary entries, (1) also gives a written incidence proof for $n = 120, 176, 253$, while the 105 entry is certified by a complete row reduction.

This proves Lemmas 1 and 2, including the full ranks at $p = 11, 23$ in Lemma 2. The paragraph preceding Lemma 1 says that the only nonternary exception to full rank or codimension one occurs for $(n, p) = (253, 5)$. The additional entries $(120, 11)$ and $(176, 17)$ in (3) correct that sentence. They do not change any of the ternary principal codes.

The distances used below come from the previously completed full coefficient enumerations in the exact row-reduced bases, together with the two separately completed enumerations for K and N at length 120. These are mathematical input certificates. For a basis of dimension k, the enumerator visits each of the 3^k coefficient vectors once, and basis independence makes their images all distinct codewords. The complete distributions are retained, not merely their minimum entries. The current checker compares the reconstructed bases with those exact bases. No large enumeration is repeated during assembly.

The common MacWilliams theorem [2, Theorem 7.1] then gives every dual coefficient over the integers. In particular the resulting parameters and minimum multiplicities are

code	n	k	d	A_d	$d^\perp$	$A^\perp_{d^\perp}$
C_{105}	105	20	33	770	6	15680
D_{105}	105	19	36	3360	5	84
C_{120}	120	21	30	112	6	33600
S	120	20	30	112	6	43680
K	120	15	48	1260	4	2100
N	120	16	48	1260	4	1260
C_{176}	176	21	56	924	6	184800
C_{253}	253	23	77	46	6	850080
D_{253}	253	22	97	3542	6	850080.

(4)

The dual dimension is $n - k$ in each row. Nonnegative integer coefficients and the dual total are checked, but those consistency checks alone do not certify an arbitrary proposed input distribution. The complete coefficient enumeration, its matrix basis, and its method are all retained. The two 105 distributions agree entry by entry with source Tables 1–2; those full distributions are reproduced in the data table accompanying this application. The two further 20-dimensional codes of Remark 2 use a smaller, justified search, described below.

3 The codes on 105 flags

The graph is the one fixed in Section 1. Here write $C = C_{105}$ and $D = D_{105}$.

Let A be the adjacency matrix over $F = \mathbb{F}_3$, and write

$$C = \langle \text{rows of } A \rangle, \qquad D = \langle A_i - A_1 : 1 \le i \le 105 \rangle.$$

The parameter identity gives $A^2 = 0$, so C and D are self-orthogonal. Exact row reduction gives $\dim C = 20$, $\dim D = 19$, and

$$C = D \oplus Fj, \tag{5}$$

where j is the constant vector. Both bases agree byte for byte with the separately completed enumeration. The two complete distributions give

	dim	d	minimum words	$(\dim C^\perp, d^\perp, \text{minimum words})$
C	20	33	770	$(85, 6, 15680)$
D	19	36	3360	$(86, 5, 84)$.

The complete dual distributions are checked again by the classical MacWilliams identity. No 3^{20} or 3^{19} enumeration is repeated. The retained distributions agree entry by entry with the two printed tables.

3.1 The full group and irreducibility

The literal coordinate permutations in the certificate preserve every entry of A. Their closure has order 241920, and their action has rank four. They generate a group G with a normal subgroup $N \cong \mathrm{PSL}_3(4)$ of order 20160 and an explicit complement $Q \cong D_{12}$ of order twelve. The complement consists of a diagonal projectivity, field conjugation and point–line duality. The checker derives their actual flag permutations, verifies $N \cap Q = 1$ and $NQ = G$, and checks the group identifications. Thus $G \cong L_3(4) : D_{12}$, with this action fixed.

The action on the row-reduced 19-dimensional basis of D is checked directly. The generator matrices span a matrix algebra of dimension $19^2 = 361$: the verifier evaluates independent generator words and row-reduces their flattened matrices. The common matrix-algebra criterion [2, Theorem 6.2] therefore proves absolute irreducibility of D. Consequently (5) is an internal decomposition into two simple modules, as required in Proposition 1.

The complete minimum-word set of C is obtained as four disjoint orbits of literal weight-33 words, of lengths $105, 105, 280, 280$. The accepted count 770 proves exhaustion, and also recovers Remark 1. There are 385 distinct supports. As for the 120- and 176-coordinate codes, form their bipartite coordinate–support incidence graph. Apply the common refinement bound [2, Theorem 6.4]. Successively individualizing coordinates $1, 2, 3, 4, 7, 12, 6$ gives cell sizes

$$105, 8, 3, 2, 4, 3, 4.$$

All coordinate colours then differ, so the theorem gives

$$|\operatorname{Aut}(C)| \le 105 \cdot 8 \cdot 3 \cdot 2 \cdot 4 \cdot 3 \cdot 4 = 241920.$$

Since $G \le \operatorname{Aut}(\Gamma) \le \operatorname{Aut}(C)$, equality follows. Every automorphism of D fixes j and hence preserves $D + Fj = C$. Conversely G preserves D, so

$$\operatorname{Aut}(D) = \operatorname{Aut}(C) = \operatorname{Aut}(\Gamma) = L_3(4) : D_{12}. \tag{6}$$

These are coordinate-permutation groups, not monomial groups.

Both retained distributions have exactly 3360 words of weight 36. Since $D \subset C$, all such words of C are in D. Their span is a nonzero G-submodule of the irreducible D, and is therefore D. This establishes the source's intrinsic definition of its codimension-one code, rather than identifying it only by parameters.

3.2 Uniqueness inside the permutation module

Put $M = F^{105}$. A supplied basis gives invariant prefixes of dimensions

$$0, 1, 20, 39, 84, 85, 104, 105.$$

The factor dimensions are $1, 19, 19, 45, 1, 19, 1$. All factors are simple. For the factors of dimensions one and nineteen, this is verified by the full matrix algebra as above. For the 45-dimensional factor use the rank-one alternative in that same theorem.

Here a recorded polynomial of a recorded sum of generator words gives P of rank one. The checker evaluates it and produces both bases of size 45. Hence its algebra has dimension 2025, proving simplicity over F and after extension. This is a certificate of the concrete factor, not an assumed classification of representations.

For each of the three 19-dimensional factors S_i, all linear equations

$$XP_g = (S_i)_g X$$

are solved, where P_g is the actual 105-coordinate permutation matrix. The full solution-space dimensions are $0, 1, 1$ in the displayed filtration order. Every basis solution has all rows in D, hence every homomorphism $S_i \to M$ has image in D.

Any 19-dimensional submodule of M is simple. Indeed its composition factors occur among those of M; dimension 45 is impossible, while the three available one-dimensional factors alone cannot account for dimension nineteen. Its sole factor must therefore have dimension nineteen. Such a simple submodule is isomorphic to one of the S_i, and its inclusion is the image of a corresponding homomorphism into M. The preceding complete equations force its image to be D. Thus D is the unique submodule of dimension nineteen in this permutation representation, proving the first assertion of Remark 2.

3.3 The other two twenty-dimensional modules

The certificate supplies invariant row-reduced bases for C_1, C_2, besides $C_3 = C$. Their containments are

	$D \subseteq C_i$	$j \in C_i$
C_1	no	yes
C_2	yes	no
C_3	yes	yes.

The only one-dimensional submodule of M is Fj. To see this directly, an invariant line has a spanning vector with entries in $\{0, a, -a\}$. Transitivity makes its zero set empty. A nontrivial scalar action would interchange the two nonzero values, forcing their multiplicities to be equal, contrary to the odd length 105. Thus the line is fixed and constant. Every 20-dimensional submodule has factors of dimensions 19 and one, by the full composition series above. The containments and uniqueness just proved give

$$\dim \mathrm{Soc}(C_1) = 1, \quad \dim \mathrm{Soc}(C_2) = 19, \quad \dim \mathrm{Soc}(C_3) = 20.$$

The three modules are therefore reducible and pairwise nonisomorphic as FG-modules, not just inequivalent as codes.

For their remaining code parameters, apply the common transitive-distance and minimum-set theorems [2, Theorems 8.1–8.2] to the actual transitive group and row-reduced information columns. For $(n, k) = (105, 20)$ and candidate distances $30, 28$, both the smaller-word and minimum-word tests require only messages of weight at most five. There are

$$\sum_{i=0}^{5} \binom{20}{i} 2^i = 583569$$

such ternary messages, including zero. The complete shared enumerator checks them once each. Its smallest nonzero output weights are $30, 28$, and literal witnesses attain those values. The averaging theorem therefore proves those minimum distances.

The unions of a literal witness orbit and its negative have sizes 672 and 720. They are invariant sets of minimum words. Exactly 342 and 396 of these words, respectively, have message weight at most five; these equal the complete restricted enumeration counts at the minimum weight. Thus every restricted minimum word is in the proposed invariant set. Averaging an arbitrary minimum word into that restriction proves that the sets are complete. This recovers $C_1 = [105, 20, 30]_3$ with 672 minimum words and $C_2 = [105, 20, 28]_3$ with 720 minimum words, as printed in Remark 2. It neither assumes a complete 3^{20} enumeration nor asserts that a short search is the full weight distribution.

3.4 The global and local representation claims

The source's proof of Corollary 1 does not justify its global statement that nineteen is the smallest dimension of a nontrivial irreducible F_3G-module. Literally that statement is false: G has a normal subgroup of index two, so its quotient acts nontrivially by the scalars $1, -1$ on a one-dimensional F_3-space. The certificate exhibits the index-two subgroup by adjoining the diagonal and field automorphisms to N; its order is 120960. No faithfulness hypothesis is stated in the printed claim. We prove the precise local replacement: the unique nonconstant simple submodule of dimension nineteen in this permutation module is D, and no nonconstant simple submodule has smaller dimension. No classification of all faithful irreducible representations is inferred.

The graph uniqueness input and all finite premises remain those stated above. Source comparisons to historical optimal-code tables are treated in the final section. Discovery used GRAPE and MeatAxe, but the final checks use literal generators, a full invariant filtration, matrix products and complete linear equations.

4 The graph and difference codes of length 120

For the residual incidence matrix $H = H_2$, equation (1) gives $HH^{\mathsf{T}} = I_{21}$ and $C_{120} = \operatorname{im} H^{\mathsf{T}}$. Thus the incidence map is an isometry from the point space to this code. The constant vector satisfies $H^{\mathsf{T}}e = j$ and $Hj = 40e = e$ over F, so $j \in C_{120}$. The common complement theorem proves

$$C_{120} \perp\!\!\!\oplus\, C_{120}^{\perp} = F^{120}, \qquad C_{120}^{\perp} = \overline{C}_{120}.$$

The 21-dimensional code and its 99-dimensional dual have the first length-120 parameters and counts in (4).

The supplementary subcode S in the proof of Proposition 2 is exactly

$$S = C_{120} \cap j^{\perp} = H^{\mathsf{T}}(e^{\perp}) = \langle (A_{120})_i - (A_{120})_1 : 1 \le i \le 120 \rangle. \tag{7}$$

To see the last equality, every row difference has sum zero, while the rows of A_{120} have the same nonzero sum $77 = 2$. Their span has dimension 21, and adjoining any one row to the difference span gives that whole space. The difference span therefore has dimension at least 20, and containment in the kernel of the nonzero sum functional gives equality. Under the isometry, that kernel is $e^{\perp}$. Since $e^{\mathsf{T}}e = 21 = 0$, the radical of $e^{\perp}$ is precisely Fe. Consequently

$$\operatorname{Hull}(S) = Fj. \tag{8}$$

The accepted exact difference-code basis and distribution prove $S = [120, 20, 30]_3$ and $S^{\perp} = [120, 100, 6]_3$, with respectively 112 and 43680 minimum words.

There is also a direct proof of the source's reflexive-code identity. On $C_{120} = H^{\mathsf{T}} F^{21}$ the operator $A_{120} + I$ acts as $H^{\mathsf{T}} J_{21}$ on point coordinates; on $C_{120}^{\perp} = \ker H$ it acts as the identity. Its image is therefore

$$\operatorname{im}(A_{120} + I) = Fj + C_{120}^{\perp} = S^{\perp}. \tag{9}$$

The two summands have dimensions one and 99 and meet trivially because C_{120} is LCD. Thus (9) is an equality of the actual codes, not a match of parameters.

4.1 Full automorphism groups of the graph and difference codes

Let Γ be the fixed length-120 graph, write $A = A_{120}$ and $C = C_{120}$, and let G be the supplied group on its coordinates.

Theorem 1. *For these actual coordinate spaces,*

$$\operatorname{Aut}(C) = \operatorname{Aut}(S) = \operatorname{Aut}(\Gamma) = G, \qquad |G| = 80640.$$

Here G is the group on the literal four coordinate generators of the retained degree-120 module certificate. Its previously checked abstract identification is $L_3(4) : 2^2$. The equality and order in this theorem have the separate elementary certificate below; the abstract identification uses that earlier exact group certificate.

Lemma 2 (Recovering a graph from its code)**.** *For this matrix, $E = J - A$ is the orthogonal projector onto C. Consequently every coordinate permutation preserving C preserves A.*

Proof. The strongly regular parameters give $A^2 = 33I + 8A + 44J$, hence $A^2 = -A - J$ over $\mathbb{F}_3$. Also $AJ = JA = 2J$ and $J^2 = 0$. Therefore

$$E^2 = J^2 - JA - AJ + A^2 = J - A = E, \qquad EA = A.$$

Every row of J lies in C: the sum of the rows of A is $77\mathbf{1} = 2\mathbf{1}$. Thus $\operatorname{im} E \subseteq C$, and $EA = A$ gives the reverse inclusion (using symmetry to identify row and column spaces). Since E is symmetric and idempotent, its kernel is $C^{\perp}$ and its image is C, so $C \cap C^{\perp} = 0$. This determines the projector uniquely. A coordinate permutation preserving C also preserves the standard dot product and hence $C^{\perp}$; conjugation must fix E. It fixes J, so it fixes $A = J - E$. □

Lemma 3 (Recovering the graph from the difference code)**.** *The collection of supports of the minimum words of S determines A.*

Proof. The literal generators of G preserve A, hence preserve both C and S. Indeed, a permuted row difference is again a difference of two rows. The exact replay verifies a supplied word w lies in the actual S and has weight 30. Its orbit under these generators consists of 112 distinct words. The complete distribution in (4) gives exactly 112 minimum words, so this orbit is all the minimum words of S. It is closed under negation and has exactly 56 distinct supports. No orbit size is substituted for an unproved completeness assertion: that assertion uses the retained exhaustive count.

Let $\mathcal{B}$ be these 56 supports and put

$$t_{ij} = |\{B \in \mathcal{B} : i, j \in B\}|.$$

The replay constructs every orbit word from w, then computes every entry of this integer incidence Gram matrix. It gives $t_{ii} = 14$, and, on the 7140 unordered distinct pairs, the complete table is

A_{ij}	t_{ij}	Number of pairs	Interpretation
0	4	2520	nonedge
1	2	3360	edge
1	6	1260	edge

Thus, for $i \neq j$, $A_{ij} = 1$ if and only if $t_{ij} \neq 4$. Any coordinate permutation preserving S preserves Hamming weight and therefore permutes $\mathcal{B}$; it preserves every t_{ij} and hence A. □

Lemma 4 (An elementary full graph bound)**.** *The retained four permutations generate all of* $\mathrm{Aut}(\Gamma)$.

Proof. First the replay verifies that each is a permutation of all 120 coordinates preserving every adjacency entry. Closing the generated permutations under composition gives exactly 80640 distinct elements; only exact finite permutation operations are used.

For the reverse inequality, apply the common refinement theorem [2, Theorem 6.4], using the deterministic choice of the first vertex in the smallest nonsingleton cell (lexicographic ties). The following are the actual chosen vertices and sizes of their cells just before individualization; vertices in this table are one-based.

Step	Chosen vertex	Cell size	Previously fixed vertices
1	1	120	none
2	5	42	1
3	2	8	1,5
4	3	2	1,5,2

After the last refinement all 120 cells are singletons. The common theorem gives $|\mathrm{Aut}(\Gamma)| \leq 120 \cdot 42 \cdot 8 \cdot 2 = 80640$. This matches the subgroup lower bound. □

Proof of Theorem 1. Every graph automorphism permutes rows of A and therefore preserves C and S. Lemmas 2 and 3 give the reverse inclusions. Lemma 4 identifies the common permutation group exactly. □

5 The socle on 120 Fano subplanes

Use the fixed $S(4, 7, 23)$ already constructed and checked by `steiner_construction.py`. Delete two named points. The 120 heptads avoiding them form a set $\mathcal{H}$. The 21 heptads containing both points, with those two removed, form the lines of a projective plane on 21 points. Its lines have size five; every two remaining points lie on one line by the Steiner property. Two such lines meet in one point, since the original heptads meet in one or three points and already share the deleted pair.

For $H \in \mathcal{H}$, every two points of H lie on a plane line meeting H in three points. Indeed the heptad defining that line shares at least two points with H and hence exactly three. These triples form the Fano subplane on the seven points of H. Let $\mathcal{F}$ be the 105 incident point–line flags of the plane, and define a 105×120 matrix R by

$$R_{(p,\ell),H} = 1 \quad \Longleftrightarrow \quad p \in H \ \text{ and } \ |\ell \cap H| = 3.$$

Thus each column records the 21 flags of the corresponding Fano subplane. All following code spaces and matrices are over $\mathbb{F}_3$.

The common square-zero incidence lemma [2, Lemma 6.1] will give self-orthogonality after the following integer table is checked.

For the fixed family, the exact integer scalar products of incidence rows are as follows. This table is checked for every pair of the 105 flags by the retained complete construction, not inferred from a sampled orbit.

relation between the two flags	$(RR^{\mathsf{T}})_{f,g}$
$f = g$	24
same point or same line, but distinct	12
distinct points and lines, with one cross incidence	6
distinct points and lines, with neither cross incidence	3.

Two distinct points and distinct lines cannot have both cross incidences. Thus the table exhausts all cases and proves $RR^{\mathsf{T}} = 0$ over $\mathbb{F}_3$. Each row has sum 24, so $Rj = 0$, and the constant vector on 120 coordinates has norm zero. Set

$$K = \mathrm{im}(R^{\mathsf{T}}R), \qquad N = K + \mathbb{F}_3 j.$$

The exact row reductions give $\dim K = 15$ and $\dim N = 16$. Consequently the sum defining N is direct. Here the direct sum is an internal decomposition of modules on the same 120 coordinates, not a concatenation of codes on disjoint coordinate sets. That common lemma proves self-orthogonality of both codes. These ranks and the full table, including its underlying block lists, are obligations of the complete finite certificate.

5.1 Recognizing the socle

The literal permutations in `module_data.json` preserve the residual Steiner graph and generate a primitive permutation group $G \cong L_3(4)$ on $\mathcal{H}$, with point stabilizer $L_3(2)$. The certificate checks primitivity, the isomorphisms and stabilizer, so this is the specified degree-120 coset action in the source. It also checks invariance of K and N. We apply the common full-matrix-algebra and socle criteria [2, Theorems 6.2–6.3]; the whole filtration and all intertwiner spaces must be checked.

Here $V = \mathbb{F}_3^{120}$. The supplied basis certifies invariant prefixes of dimensions

$$0, 15, 34, 49, 64, 65, 84, 99, 100, 119, 120.$$

The ten factor dimensions and homomorphism-space dimensions are

i	1	2	3	4	5	6	7	8	9	10
$\dim D_i$	15	19	15	15	1	19	15	1	19	1
$\dim \mathrm{Hom}_G(D_i, V)$	1	0	0	0	1	0	1	1	0	1.

Triangularity of the generator matrices verifies invariance of all prefixes. For each factor the matrix algebra has dimension $(\dim D_i)^2$, proving simplicity by the common criterion. The first prefix is exactly K. For each factor, the verifier forms *all* equations $XP_g = D_g X$ for the given generators and computes their full solution space. Every row of every basis solution belongs to N. Linearity then establishes the hypothesis for every homomorphism, not only a selected map. The constant line is simple and K is the first simple factor, so the common criterion yields

$$\mathrm{Soc}(\mathbb{F}_3^{120}) = N = K \oplus \mathbb{F}_3 j.$$

This calculation does not classify every irreducible module of $L_3(4)$. The source's classification assertion concerning three nonisomorphic faithful irreducible modules of dimension 15 is retained from its reference [13], Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). Our certificate identifies a particular such simple module; its faithfulness also follows because G is simple and the representation is nontrivial.

5.2 Weights and full permutation automorphisms

Complete enumeration in the certified bases gives

code	number of words	dimension	d	words of weight d	$d^\perp$
K	14348907	15	48	1260	4
N	43046721	16	48	1260	4.

The enumerator visits each of the 3^k coefficient vectors exactly once; independence of the basis makes these distinct words. It records the full weight distribution. The complete dual distributions follow from the common MacWilliams theorem [2, Theorem 7.1]. In both cases the dual coefficients at weights $1, 2, 3$ vanish. At weight 4 they are 2100 for $K^\perp$ and 1260 for $N^\perp$. Thus the dual dimensions are 105 and 104, respectively, and the stated dual minimum distances follow. These are complete finite computations, not distance bounds inferred from self-orthogonality.

The orbit under G of the supplied weight-48 word contains 1260 distinct words and 630 distinct supports. Since $K \subset N$ and both codes have exactly 1260 minimum words, this orbit is the complete minimum-word set of both codes. Every coordinate-permutation automorphism of either code therefore acts on its support-incidence structure, with 120 coordinate vertices and 630 distinct support vertices.

The common refinement theorem [2, Theorem 6.4] bounds the full automorphism group of this structure. Successively individualize the coordinate vertices $1, 6, 48, 10, 2, 3$ in the literal certificate ordering. Their successive refined colour classes have sizes

$$120, 21, 2, 4, 2, 2.$$

After these choices all coordinate classes are singletons. The theorem gives the upper bound $120 \cdot 21 \cdot 2 \cdot 4 \cdot 2 \cdot 2 = 80640$.

The explicit permutations supplied in the certificate generate a group A of order 80640 and preserve each of K, N. Thus both code automorphism groups attain the upper bound and equal A. The certificate checks $A' = G$ and supplies two commuting involutions generating a complement of order 4, disjoint from G. Consequently

$$\mathrm{Aut}(K) = \mathrm{Aut}(N) = A \cong L_3(4) : 2^2.$$

No graph-isomorphism program is needed for this upper bound. Such software was useful in discovering the permutations, but the retained proof checks literal permutations, ordinary group orders and the full refinement chain.

6 The irreducible code on 176 residual blocks

Here $H = H_1$, $C = C_{176} = H^{\mathsf{T}}(e^\perp)$ and $\dim C = 21$ by (1). The integral Gram matrix is $40I_{22} + 16J_{22}$, reducing to $I_{22} + J_{22}$. The common incidence theorem proves LCD and the two complement–dual identities. The constant vector belongs to $C^\perp$, because its image under H is $56e$, which annihilates $e^\perp$. Thus the full orthogonal decomposition in the source is established before any module classification.

6.1 The exact module and group certificate

The literal permutations in `module_data.json` preserve every adjacency entry of Γ. Their closure, computed by exact permutation composition, has order 443520. They also permute the 22 point-incidence rows of H. The resulting faithful degree-22 action is explicitly identified with `MathieuGroup(22)` by an isomorphism checked in GAP. Thus the subgroup G they

generate is M_{22}. This conventional group identification is distinct from proving that G is the *full* code automorphism group. The latter will follow from the common refinement theorem [2, Theorem 6.4]. The auxiliary group calculation also verifies that the degree-176 action is primitive with stabilizer A_7 of order 2520.

The row-reduced adjacency basis is checked byte for byte against the basis of the retained complete enumeration. For each coordinate generator, its images on these 21 rows give a 21×21 representation matrix. The checker reconstructs those images from the proposed matrices, so the matrix module is the actual code C, not a representation with only the same dimension. Deterministic multiplication and elimination produce 441 independent generator words, all of length at most eight. Each word is evaluated again and the 441 flattened matrices are row-reduced. Their rank is 441. A separate GAP calculation confirms the same generated algebra dimension. The common matrix-algebra theorem [2, Theorem 6.2] proves that C is absolutely irreducible. This action is faithful as well: the 176 columns of a code basis are distinct, so a coordinate permutation fixing every codeword fixes every coordinate. Column distinctness is checked in the certificate.

The retained complete enumeration gives $d(C) = 56$ and exactly 924 words of weight 56. The supplied literal weight-56 word belongs to C. Its orbit under G has 462 words and is disjoint from the orbit of its negative. Together those two orbits have 924 words, so they exhaust the minimum words. Their supports are the same 462 distinct subsets. These assertions are checked by finite orbits and the already accepted count; no new exhaustive codeword computation is involved.

For this support-incidence graph, with $176+462$ vertices, the deterministic refinement rule of the common refinement theorem [2, Theorem 6.4] successively individualizes coordinates $1, 5, 29, 61, 2$. The corresponding cell sizes are

$$176, \quad 70, \quad 6, \quad 2, \quad 3.$$

All coordinate colours then differ. Therefore

$$|\operatorname{Aut}(C)| \le 176 \cdot 70 \cdot 6 \cdot 2 \cdot 3 = 443520.$$

Every graph automorphism preserves its adjacency row space, so $G \le \operatorname{Aut}(\Gamma) \le \operatorname{Aut}(C)$. The already checked order of G forces equality throughout:

$$\operatorname{Aut}(C) = \operatorname{Aut}(\Gamma) = G \cong M_{22}. \tag{10}$$

This upper bound uses no graph-isomorphism backend. Such software was used only to discover the two literal generators.

The complete retained weight distribution determines the dual by the common MacWilliams theorem [2, Theorem 7.1]. The certificate computes all coefficients exactly, checks nonnegative integrality and total 3^{155}, and obtains zero at weights one through five and 184800 at weight six. Thus the parameters are $[176, 21, 56]_3$ and $[176, 155, 6]_3$, with the asserted minimum counts.

6.2 The precise representation boundary

Theorem 1(c) describes the *unique* 21-*dimensional* irreducible F_3-module for M_{22}. Proposition 4's final sentence omits the dimension, but its proof immediately restores it and invokes the known 3-modular character table. We retain that intended classification statement, and the proof's further assertion that 21 is the smallest nontrivial irreducible dimension, from the source's reference [13]: Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). These assertions are understood up to module isomorphism. Our full matrix-algebra calculation proves irreducibility of this particular module; it does not classify all modules. The

unqualified literal uniqueness sentence cannot be correct, since the trivial one-dimensional module is irreducible too.

The source's historical near-optimality comparison for the dual is a separate assertion based on its tables, not a current table claim or a consequence of (10). No additional irreducibility is asserted for the 155-dimensional dual, which contains the constant line. The group in (10) consists of coordinate permutations; allowing scalar monomial maps would change the automorphism convention.

7 The augmentation code on 253 heptads

Let $\mathcal{D}$ be the specified $S(4, 7, 23)$, with point set Ω and 253 heptads. Its point–block incidence matrix is H. Two distinct heptads intersect in one or three points, and Γ joins them in the latter case. Write A for its adjacency matrix. All vector spaces in this note are over $F = \mathbb{F}_3$. Put

$$E = \{x \in F^{23} : \textstyle\sum_i x_i = 0\}, \quad C = \operatorname{im} A, \quad D = H^{\mathsf{T}} E.$$

The shared construction checks the Steiner condition and actual incidence matrix, not merely the strongly regular parameters.

Theorem 5. *The code C is the orthogonal direct sum $\langle \mathbf{j} \rangle \oplus D$, where D is an absolutely irreducible 22-dimensional module for M_{23}. Both C and D are LCD. Their parameters and minimum multiplicities are*

code	*parameters*	*minimum words*
C	$[253, 23, 77]_3$	46
D	$[253, 22, 97]_3$	3542
$C^{\perp}$	$[253, 230, 6]_3$	850080
$D^{\perp}$	$[253, 231, 6]_3$	850080.

The full groups of coordinate permutations preserving C and Γ are equal and isomorphic to M_{23}.

Proof. The common residual-incidence theorem [2, Theorem 2.2] applies to the fixed system and gives $HH^{\mathsf{T}} = 2I_{23}$, $A = H^{\mathsf{T}}(J_{23} - I_{23})H$, $C = \operatorname{im} H^{\mathsf{T}}$ and the orthogonal splitting $C = Fj \perp \oplus H^{\mathsf{T}}(e^{\perp})$. The two point-space summands are nondegenerate because $23 \neq 0$ in F. Thus it proves both LCD statements and the displayed dimensions.

We next prove the irreducibility, using explicit matrices in place of a modular character-table inference. The five permutations of Ω in `generators.json` preserve the full heptad set. They therefore act on the code coordinates and on E. In the basis $e_i - e_{23}$, $1 \leq i \leq 22$, the row matrix for a point permutation g has entries

$$M(g)_{ij} = \mathbf{1}_{j=g(i)} - \mathbf{1}_{j=g(23)}, \qquad 1 \leq i, j \leq 22.$$

An indicator with point 23 is zero because $j \leq 22$. The certificate checks that multiplication by these matrices agrees with the induced permutation of all 253 coordinates on the actual rows $h_i - h_{23}$. It produces 484 independent words in these matrices, each of length at most six, and reevaluates every word before reducing the flattened matrices. They span $M_{22}(F)$.

The common full-matrix-algebra criterion [2, Theorem 6.2] proves absolute irreducibility of E and its equivariant image D.

The existing enumeration records contain the complete weights of the row-reduced adjacency code and of the difference code. Their bases are compared byte for byte with the bases reconstructed here. The latter row space is D: its dimension is 22, it lies in C and is orthogonal to $\mathbf{j}$. The recorded enumerations have respectively 3^{23} and 3^{22} coefficient vectors. They

give the two distances and counts in the first two rows of the table. The dual distributions follow from the common MacWilliams theorem [2, Theorem 7.1]. All coefficients are evaluated over the integers. Nonnegativity, integrality and total 3^{253-k} are checked. Both duals have no words of weights one through five and exactly 850080 of weight six. This proves the table without repeating either exhaustive enumeration.

The 46 words $h_i, -h_i$ are precisely the minimum words, by their weights and the complete count. The common Steiner recognition proposition [2, Proposition 2.3] therefore identifies $\mathrm{Aut}(C)$ with the block action of $\mathrm{Aut}(\mathcal{D})$. This explicitly discharges its minimum-word hypothesis.

For an explicit upper bound, form the bipartite point–block incidence graph, with its two parts initially given different colours. Refine the colours by the old colour and the counts of neighbours of each colour. After individualizing points $1, 2, 3, 4, 5, 6$ successively, the certificate obtains current cell sizes

$$23, \quad 22, \quad 21, \quad 20, \quad 16, \quad 3.$$

All points are then distinguished. A pointwise automorphism fixes every heptad, since distinct heptads have different point neighbourhoods. The common refinement theorem [2, Theorem 6.4] consequently bounds $|\mathrm{Aut}(\mathcal{D})|$ by $23 \cdot 22 \cdot 21 \cdot 20 \cdot 16 \cdot 3 = 10200960$. Exact group computations on the literal generators give a subgroup of this order and an isomorphism to the standard group M_{23}. The induced block action is checked to be faithful. Thus the upper bound is attained and identifies the full design and code groups. Finally the subgroup preserves Γ, and every graph automorphism preserves its adjacency row space. Therefore $G \leq \mathrm{Aut}(\Gamma) \leq \mathrm{Aut}(C) = G$, proving the asserted equality. The upper bound uses only incidence refinement; graph-isomorphism software was used to discover the generators and is not a final premise. □

Qualifications to Proposition 5. The main Theorem 1(d) is recovered as stated. Proposition 5 additionally assigns distance eight to $D^{\perp}$ and declares it irreducible. Both assertions are false for these codes. Since $D \subset C$, the invariant space $C^{\perp}$ of dimension 230 is a proper nonzero subspace of $D^{\perp}$, whose dimension is 231. The same inclusion already gives $d(D^{\perp}) \leq d(C^{\perp}) = 6$; the exact value six is verified above. The proposition also calls D the "unique and smallest" irreducible module without qualifications. The unqualified statement overlooks the trivial one-dimensional module. Its intended classification requires an input separate from our matrix certificate. We retain the complete 3-modular character table of M_{23}, as in the source's reference [13], Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). Its irreducible degrees over an algebraic closure are

$$1, 22, 45, 45, 104, 104, 231, 253, 770, 770, 990, 990, 1035.$$

The accompanying GAP calculation checks these entries against its installed table `M23mod3`; it does not reprove completeness of that table. Together with standard scalar extension and descent for simple modules over finite fields, the table gives a unique 22-dimensional irreducible F-module, and 22 is the least nontrivial degree. Indeed a simple module splits after scalar extension into an orbit of absolutely simple constituents of equal degree. Below degree 22 the only available constituent is trivial, which forces the original simple module to be trivial. At degree 22 the unique constituent is the one already realized by D; descent of module isomorphisms gives uniqueness over F. These are retained classification premises, explicitly distinct from the direct proof of irreducibility of D. The source's open uniqueness question for the strongly regular parameters $(253, 140, 87, 65)$ is not resolved here.

8 The source questions and precise qualifications

Section 6 asks whether equality of a graph code with the orthogonal dual of its complementary graph code forces LCD, and whether the converse holds. The forward direction holds over every field: that equality gives $A(J - I - A) = 0$, and the common projection theorem [2, Theorem 1.2] gives LCD. Its conclusion also identifies the exact one-dimensional defect when both degrees vanish in the field.

The converse fails already for the four-cycle over $\mathbb{F}_3$. In cyclic coordinate order its code is spanned by $(1, 0, 1, 0)$ and $(0, 1, 0, 1)$. Their Gram matrix is $2I_2$, so this code is LCD. Its complementary graph is two disjoint edges, whose adjacency matrix is invertible. The complementary code is the whole space, whose dual is zero, and hence cannot equal the two-dimensional cycle code. The examples of lengths 120 and 176 still give the affirmative existence answer sought in the source's introduction: they go beyond triangular graphs and have exact complementary dual codes. These arguments make no claim about the historical priority of either implication.

The uniqueness problem for strongly regular parameters $(253, 140, 87, 65)$, mentioned in Section 6.3, remains outside our result. Our 253-coordinate code is constructed from the specified Steiner system, and no uniqueness of every graph with those parameters is used. Remark 3's discussion of locating the other two 15-dimensional irreducible modules is also not a construction of those modules here. Their abstract existence and classification remain the explicitly retained Brauer-table input; the specified degree-120 permutation module and its particular socle are proved above.

The asymptotic Gilbert–Varshamov result of Sendrier [5] concerns families of LCD codes. Membership of one finite code in the LCD class does not by itself assert a finite optimality comparison or an asymptotic property of that single code. We retain the cited asymptotic result with that meaning. Likewise, the comparisons to code tables in Propositions 1–4 and Corollary 1 are historical comparisons, not updated optimality claims. None is used to establish a dimension, distance or full automorphism group here.

The literal and intended readings of the representation statements have been separated throughout. The 105 global least-degree statement is false without an additional restriction, as witnessed by the actual nontrivial one-dimensional quotient character. The dimension qualifier in the 176 main theorem is necessary. For length 253 the printed 231-dimensional dual has distance six rather than eight and is reducible; its invariant 230-dimensional subcode already disproves irreducibility. The main Theorem 1(d), which does not assert those erroneous dual claims, is preserved. A matrix-algebra certificate proves irreducibility of a specified representation; global uniqueness and minimum-degree classifications require the separately retained table inputs.

Computations and formal scope. The finite premises are the complete specified constructions, literal matrices and group generators, full filtration and intertwiner systems, complete weight certificates, and support-refinement upper bounds. Their executable files and mathematical data are inventoried in full. Discovery with graph-isomorphism software or MeatAxe is not a replacement for these final checks. Exact arithmetic in the declared Python, C and GAP implementations is trusted in the finite certificates. Operational drivers and optional weakening tests are listed separately. The nine previously checked generic Lean declarations for projection and averaging retain their precise restricted scope; this application is not an end-to-end Lean formalization. No new source, question, historical novelty, or compression credit is assigned by this author draft.

A The two complete distributions on 105 coordinates

This table reproduces both nonzero supports of the weight distributions in source Tables 1–2, using the exact accepted enumeration. All omitted weights have multiplicity zero. The zero word is included. The totals are 3^{20} and 3^{19}, respectively.

Weight	C_{105}	D_{105}
0	1	1
33	770	0
36	3360	3360
42	29640	9480
45	38304	336
48	141120	100800
51	987840	144480
54	3649800	1659840
57	21993720	6552000
60	108522288	36809136
63	303515520	101499720
66	606412800	202063680
69	818230000	269951640
72	804875190	272891220
75	526684032	172986912
78	222579280	74177040
81	58508240	20054720
84	9106500	2743020
87	1230880	544320
90	127008	54432
93	128520	0
96	15680	15120
99	3360	0
105	548	210

References

[1] D. Leemans and B. G. Rodrigues, Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph, *Math. Commun.* 21 (2016), 239–249.

[2] *Incidence operators, orthogonal codes and permutation modules*, fixed common manuscript, additive edition 2; theorem labels and exact edition are bound in the accompanying application manifest.

[3] K. Coolsaet, The uniqueness of the strongly regular graph $\mathrm{srg}(105, 32, 4, 12)$, *Bull. Belg. Math. Soc. Simon Stevin* 12 (2005), 707–718, Theorem 3; retained as in [1, reference 6].

[4] C. Jansen, K. Lux, R. Parker and R. Wilson, *An Atlas of Brauer Characters*, Clarendon Press, Oxford, 1995; retained classification inputs as in [1, reference 13].

[5] N. Sendrier, Linear codes with complementary duals meet the Gilbert–Varshamov bound, *Discrete Math.* 285 (2004), 345–347; retained asymptotic statement.